

Problem Sheet 9 Solutions

1.

The RMS thermal noise voltage is

$$V_{TH} = \sqrt{4k_B T R B}$$

$$k_B = 1.38 \times 10^{-23} \text{ J/K}$$

$$R = 10^6 \Omega$$

$$T = 298 \text{ K}$$

$$B = 1000 \text{ Hz}$$

$$V_{TH} = 4 \mu\text{V}_{\text{RMS}}$$

2.

(a) The RMS shot-noise current is

$$i_n = \sqrt{2eIB}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

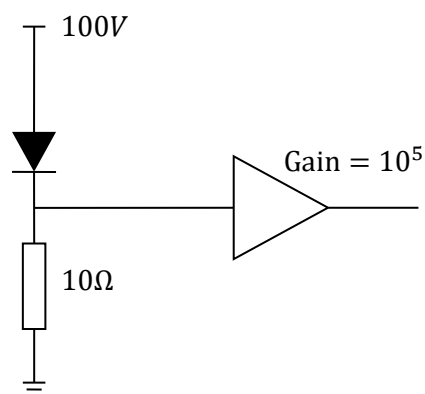
$$I = \frac{V - 0.6}{R} = 9.94 \text{ A}$$

$$B = 10^5 \text{ Hz}$$

$$i_n = 5.64 \times 10^{-7} \text{ A}_{\text{RMS}}$$

(b) The noise voltage generated across the resistor is $i_n R = 5.64 \mu\text{V}_{\text{RMS}}$.

For an amplifier with voltage gain 10^5 the noise voltage at the output will be $0.56 \text{ V}_{\text{RMS}}$.



(c) The RMS thermal noise voltage across the resistor is $\sqrt{4k_B T R B}$ which is $1.28 \times 10^{-7} \text{ V}_{\text{RMS}}$ at $T = 298 \text{ K}$, so at the output of the amplifier we will have only $0.013 \text{ V}_{\text{RMS}}$ due to thermal noise.

Instrumentation

3.

(a) A noise factor of 1 means that noise power out = noise power in (this amplifier has unity gain). This also means that the amplifier adds no additional noise to the signal. This is impossible in any real-world device, since any physical amplifier will have resistors (generating thermal noise) and semiconductor junctions (generating shot noise). However, a carefully-designed low-noise amplifier can indeed have a very low noise figure, even if the 'perfect' amplifier is impossible.

(b) For shot > thermal

$$\begin{aligned} R_{in}\sqrt{2eIB} &> \sqrt{4kTR_{in}B} \\ R_{in}eI &> 2kT \\ V_R &> \frac{2kT}{e} \end{aligned}$$

4.

(a) Bandwidth $B = 10^5 - 100 = 99900$ Hz.

The noise factor F relates the signal-to-noise ratios at the input and output of the amplifier:

$$\frac{\text{SNR}_{\text{in}}}{\text{SNR}_{\text{out}}} = 4.$$

If the input signal is represented by the RMS voltage V_{s_i} and the RMS noise voltage at the input is V_{n_i} , then we can write

$$\begin{aligned} \frac{V_{s_i}^2}{V_{n_i}^2} &= \frac{4V_{s_o}^2}{V_{n_o}^2} \\ \frac{V_{s_i}}{V_{n_i}} &= \frac{2V_{s_o}}{V_{n_o}} \end{aligned}$$

$$\text{SNR}_{\text{out}} = 20\text{dB} = 20 \log\left(\frac{V_{s_o}}{V_{n_o}}\right)$$

$$V_{s_o} = V_{n_o} \times 10,$$

and, therefore,

$$\frac{V_{s_i}}{V_{n_i}} = 20$$

$$\frac{I_{in}R_{in}}{\sqrt{4kTRB}} = 20$$

$$I_{in} = \frac{20}{10^4} (4 \times 1.4 \times 10^{-23} \times 298 \times 10^4 \times 99900)^{0.5}$$

$$I_{in} = 8.2 \text{ nA}_{\text{RMS}}$$

(b)

$$\text{SNR}_{\text{out}} = 40\text{dB}$$

$$V_{s_o} = 100V_{n_o}$$

$$\frac{V_{s_i}}{V_{n_i}} = 200 = \frac{I_{in}R_{in}}{\sqrt{4kTRB}}$$

$$T = 3\text{K}$$

5.

(a) Noise power $\propto 1/f$. So, as the frequency f of a measurement $\rightarrow 0$ then the noise power $\rightarrow \infty$. However, the extreme low-end frequency measurements are constrained by the length of time over which the measurement has been made. To make a truly DC measurement we would have to observe that our measureable has been constant for a very long period of time (ultimately an infinite length of time for a measurement at 0 Hz). So, there is no such thing as a truly DC measurement, only a very low frequency measurement. The noise per unit frequency tends to infinity; however, our actual low-frequency limit is constrained by the duration of the measurement, and is ultimately bounded. The numerical example in part (b) should help to make this clearer.

(b) Integrated noise power

$$N_f = k \int_{f_l}^{f_h} \frac{df}{f} = k \ln \frac{f_h}{f_l},$$

In any decade of frequency the noise is

$$N_f = 2.3k.$$

In our amplifier the upper frequency limit of the bandwidth is 10 kHz.

For the amplifier which has been on for 1 day, the lower frequency limit is $\sim 10^{-5}$ Hz. The bandwidth spans approximately 9 decades. The amplifier on for 100 days has a bandwidth of approximately 11 decades. So,

$$\frac{P_{100}}{P_1} = \frac{11 \times 2.3k}{9 \times 2.3k} = 1.22.$$